

Input parameters

Item	Value	Unit	Formula	Source
Positive electrode thickness	50	um	design override	bridge/params_v8.json
Negative electrode thickness	60	um	design override	bridge/params_v8.json
Separator thickness	9	um	design override	bridge/params_v8.json
Positive particle radius	1.2	um	design override	bridge/params_v8.json
Negative particle radius	2	um	design override	bridge/params_v8.json
Positive porosity	0.4		design override	bridge/params_v8.json
Negative porosity	0.42		design override	bridge/params_v8.json
Positive AMVF	0.58		design override	bridge/params_v8.json
Negative AMVF	0.56		design override	bridge/params_v8.json
Separator porosity	0.55		design override	bridge/params_v8.json
Electrolyte conductivity	1.4	S/m	design override (estimate class)	bridge/params_v8.json
Cation transference number	0.5		design override (estimate class)	bridge/params_v8.json
Cooling heat transfer coefficient	45	W/m2 K	design override (forced air)	bridge/params_v8.json
Rated (nominal) capacity	3	Ah	cell rating = designed capacity	bridge/params_v8.json
Positive density	3262	kg/m3	base set	Chen2020
Negative density	1657	kg/m3	base set	Chen2020
Separator density	397	kg/m3	base set	Chen2020
Positive c_max	63104	mol/m 3	base set	Chen2020
Negative c_max	33133	mol/m 3	base set	Chen2020
Positive solid diffusivity	4e-15	m2/s	base set (excluded design lever)	Chen2020
Negative solid diffusivity	3.3e-14	m2/s	base set (excluded design lever)	Chen2020
Electrode area	0.1027	m2	height x width	Chen2020 (0.065 x 1.58)

Item	Value	Unit	Formula	Source
Voltage window	2.5 - 4.2	V	base set cut-offs	Chen2020

Capacity and energy

Item	Value	Unit	Formula	Source
1C discharge capacity	2.66830271 6049212	Ah	simulation integration	cell/r4_v8_1c_dfn.json:capacity_a h
5C discharge capacity	2.62613110 189783	Ah	simulation integration	cell/r4_v8_5c_dfn.json:capacity_a h
5C retention	0.98419534 11441177		5C capacity / 1C capacity	cell/r4_v8_derived.json:rate_retent ion_5c
Discharge energy	9.83292707 094455	Wh	time integration of V*I	cell/r4_v8_energy.json:energy_wh
Midpoint voltage	3.92150480 5139366	V	simulation	cell/r4_v8_energy.json
DC resistance (10% discharge)	0.00182867 2606044949	ohm	(start OCV - V@10%) / I_1C	cell/r4_v8_energy.json:dc_r_ohm
Open-circuit voltage (contract)	4.11222721 6955182	V	$\sqrt{\text{PD} \times 4 \times \text{DCR} \times \text{mass}}$	derived from cell/r4_v8_energy.json
Peak power	2311.84256 656537	W	PD x mass	derived

Energy density

Item	Value	Unit	Formula	Source
Energy	9.83292707 094455	Wh	simulation integration	cell/r4_v8_energy.json:energy_wh
Cell mass (contract)	0.03161634 3915	kg	sum layer thickness x area x (1-porosity) x density; electrolyte excluded	cell/r4_v8_energy.json:mass_kg
Cell volume (stack)	1.50969e-0 5	m3	area x stack thickness	cell/r4_v8_energy.json:volume_m3
Gravimetric energy density	311.007721 1134913	Wh/kg	energy / mass	cell/r4_v8_energy.json
Volumetric energy density	651.320938 135945	Wh/L	energy / volume	cell/r4_v8_energy.json

Item	Value	Unit	Formula	Source
Power density	73121.7554 0539157	W/kg	$V_{OC}^2 / (4 \times DCR) / \text{mass (contract)}$	cell/r4_v8_energy.json:power_density_w_kg

N-P and mass

Item	Value	Unit	Formula	Source
Negative areal capacity (full-lithiation)	29.8371500 4666667	Ah/m ²	$c_{max} \times F / 3600 \times L \times AMVF$	derived from Chen2020 + params_v8.json
Positive areal capacity (full-lithiation)	49.0469704 8888889	Ah/m ²	$c_{max} \times F / 3600 \times L \times AMVF$	derived from Chen2020 + params_v8.json
N/P ratio	0.60833828 77526754		neg capacity density x thickness / pos capacity density x thickness	deliverable formula
Charged-state negative stoichiometry	0.90139739 83641687		initial concentration / c_{max}	Chen2020
Charged-state positive stoichiometry	0.26999873 22515213		initial concentration / c_{max}	Chen2020
Layer mass positive coating	0.01005022 2	kg	layer_kg_m2 x area	cell/r4_v8_energy.json
Layer mass negative coating	0.00592205 1720000002	kg	layer_kg_m2 x area	cell/r4_v8_energy.json
Layer mass Al CC	0.00443664	kg	layer_kg_m2 x area	cell/r4_v8_energy.json
Layer mass Cu CC	0.01104230 4	kg	layer_kg_m2 x area	cell/r4_v8_energy.json
Layer mass separator	0.00016512 6195	kg	layer_kg_m2 x area	cell/r4_v8_energy.json
Anode-limited balance check	2.9907 x 0.56/0.63 = 2.658 vs 2.6683 measured		1C capacity scales with negative AMVF (V6 -> V8)	r4_v6_1c_dfn.json + r4_v8_1c_dfn.json

Process parameters

Item	Value	Unit	Formula	Source
Areal density positive	0.09786	g/m ²	thickness x (1-porosity) x density	derived

Item	Value	Unit	Formula	Source
Areal density negative	0.05766360 000000001	g/m2	thickness x (1-porosity) x density	derived
Compaction density positive	1.9572	g/cm3	density x (1-porosity) / 1000	derived
Compaction density negative	0.96106000 000000001	g/cm3	density x (1-porosity) / 1000	derived
Electrolyte fill amount	6.18048600 000000001	g	pore volume x 1.2 g/cm3 (literature) x fill factor 1.0	derived
Formation recommendation	0.1C CC to 4.2V, 25C, 2 cycles		design recommended value; production-line tuning required	design note
Diffusion time positive	359.999999 99999999	s	r^2 / D_s	derived
Diffusion time negative	121.212121 2121212	s	r^2 / D_s	derived